

# SALES REPORT

*Performance Overview and Key Insights*



**by Aladeloye Esther Aderonke**

ALT/SOD/025/0045

## Perspective

This analysis was conducted from a sales analytics perspective, with the primary objective of evaluating revenue generation and profitability, and identifying top-performing products and territories to support decision-making. The analysis focuses on four major areas:

- Sales overview — how much was sold
- Cost and profit analysis — how much it cost and how much profit was made
- Product performance — what customers are buying
- Territory analysis — where sales are coming from

## Key Assumptions

The following assumptions were made to ensure consistency and analytical validity:

- StandardCost was assumed to represent the actual cost of goods sold, as a detailed operational cost breakdown was not available.
- Missing categorical values replaced with “Unknown” were assumed not to significantly distort customer or product segmentation.
- OrderDate was assumed to represent the point of revenue recognition and was used for trend and time-series analysis.

## Methodology

The AdventureWorks dataset, containing sales, product, customer, and HR information, was used in this project. The relevant tables required for analysis were imported into Power BI, including Sales.SalesOrderDetail, Sales.SalesOrderHeader, Production.Product, Production.ProductSubcategory, Production.ProductCategory, Sales.Customer, Person.Person, and Sales.SalesTerritory.

After importing the tables into Power BI, the following data transformations were carried out using Power Query Editor to prepare the data for analysis:

- Column data types were corrected where necessary (e.g., ID columns converted from date to text format).
- Categorical fields containing abbreviations were transformed into meaningful labels to improve interpretability — Product Class: L → Low, M → Medium, H → High; Product Line: M → Mountain, R → Road, S → Standard, T → Touring; Product Style: W → Women's, M → Men's, U → Universal.
- Columns containing null values were reviewed. Where appropriate, missing categorical values were replaced with “Unknown,” and irrelevant columns with excessive nulls were removed.
- After data cleaning, the dataset was loaded into Power BI using the Close & Apply function.
- A relational data model was constructed with relationships established via primary/foreign keys: SalesOrderDetail[ProductID] → Product[ProductID], SalesOrderHeader[CustomerID] → Customer[CustomerID], SalesOrderHeader[TerritoryID] → SalesTerritory[TerritoryID], and Product[ProductSubcategoryID] → ProductSubcategory[ProductSubcategoryID]. All relationships were configured as one-to-many.
- Core measures were created: Total Revenue, Total Cost, Total Profit, Profit Margin (%), and Total Orders — forming the analytical foundation for all dashboard visualizations.

Each analytical area was translated into an interactive dashboard page using appropriate visualizations, including KPI cards, line charts, bar charts, scatter plots, and pie/donut charts.

## Key Insights

### 1. Overall Sales Performance

The company generated total revenue of \$109.85 million, with 275,000 units sold across 31,465 orders, serving approximately 19,820 customers. Revenue and sales volume showed consistent growth from 2011 to 2013, peaking in 2013 before declining in 2014. This indicates strong expansion during the early period, followed by a slowdown that may reflect market saturation, competitive pressure, or operational inefficiencies.

Universal-style products account for the highest quantity sold (over 56%), suggesting that broad-appeal products drive volume. The scatter plot reveals a positive relationship between order size and revenue, with a concentration of smaller orders generating moderate revenue and fewer high-value, high-revenue transactions.

## **2. Cost and Profit Analysis**

Total cost stands at approximately \$100 million, resulting in total profit of \$9.37 million and an overall profit margin of 9%. While the company is profitable, this margin indicates limited efficiency in cost control or pricing strategy, suggesting a significant proportion of revenue is absorbed by production and operational costs.

Production and operational costs increased from 2011 through 2013 before declining in 2014. However, profit by year shows a sharp increase beginning in 2013, remaining strong into 2014. Profit margin by year shows recovery after a dip in 2012, reaching its highest level in 2014 — indicating the company became more efficient at converting revenue into profit even as overall sales declined, demonstrating operational efficiency gains and cost management.

## **3. Product Performance**

The Bikes category dominates revenue generation, contributing more than 86% of total revenue. Within this category, Mountain Bikes represent the most profitable subcategory. Road and Mountain bikes are the strongest contributors to overall revenue and profit, while Standard and Touring lines generate significantly lower revenue and may not be contributing proportionally to profitability. The top 10

products contribute a substantial share of total revenue, indicating a concentration of performance among a limited number of high-performing items.

#### **4. Territory Analysis**

The United States generates the highest revenue contribution (approximately \$63 million), making it the company's primary market. The Southwest territory leads in total units sold and total revenue, while Australia generates the highest profit among territories. However, the presence of negative profit in selected territories indicates uneven performance and potential operational inefficiencies across regions.

### **Recommendations**

#### **1. Improve Profit Margins Through Cost Optimization**

Given the current 9% profit margin, management should conduct a detailed cost review to identify inefficiencies in production, distribution, and regional operations. Pricing strategies for high-volume products such as Road bikes should be reassessed to improve margin performance without negatively affecting demand.

#### **2. Focus on High-Performing Product Lines**

The Mountain and Road product lines should receive priority investment in marketing, product development, and inventory management, given their strong contribution to revenue and profit. Conversely, underperforming lines such as Standard and Touring products should be reviewed to determine whether repositioning, cost reduction, or discontinuation is necessary.

#### **3. Strengthen High-Value Market Segments**

The United States and Southwest territories should remain strategic focus areas due to their strong revenue contribution. Best practices from profitable regions such as Australia should be analyzed and replicated in weaker territories. For regions experiencing losses, pricing, logistics costs, and customer demand patterns should be reviewed to restore profitability.

#### **4. Protect and Expand Top Products**

Since a small group of products contributes significantly to revenue, inventory planning and supply chain stability for these products should be prioritized, with additional marketing efforts to further increase their contribution.

#### **5. Address Revenue Decline After 2013**

The decline in revenue observed after 2013 requires investigation. Possible factors such as market competition, pricing adjustments, customer retention, or economic conditions should be analyzed to prevent further slowdown.

### **Conclusion**

Overall, the company demonstrates strong revenue generation capacity and clear product strengths, particularly within the Bikes category and Mountain product line. However, profitability margins remain modest, and regional performance is uneven.

By focusing on cost optimization, high-performing products, profitable territories, and strategic pricing improvements, the company can enhance shareholder value, increase profitability, and support sustainable long-term growth.

## SALES OVERVIEW DASHBOARD

Total Orders

31.465K

Total Quantity Sold

275K

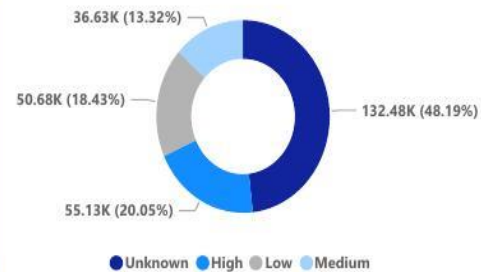
Total Revenue

109.85M

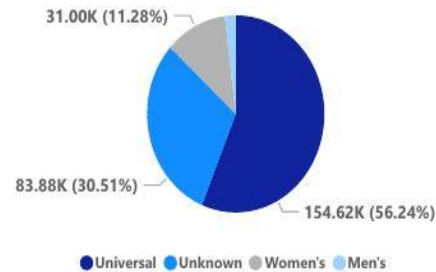
Total Customers

19.82K

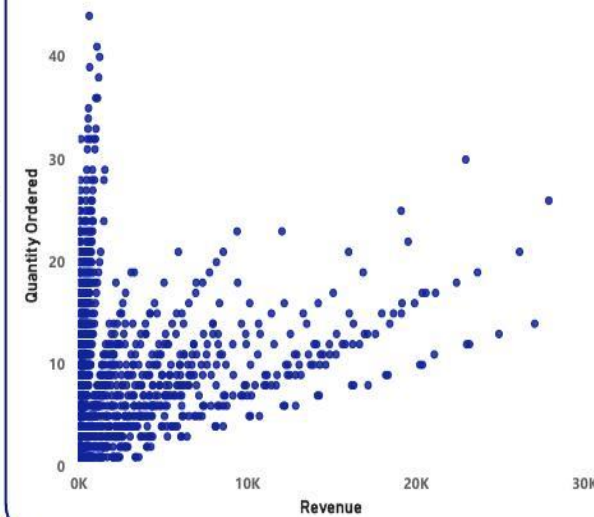
Total Quantity by Class



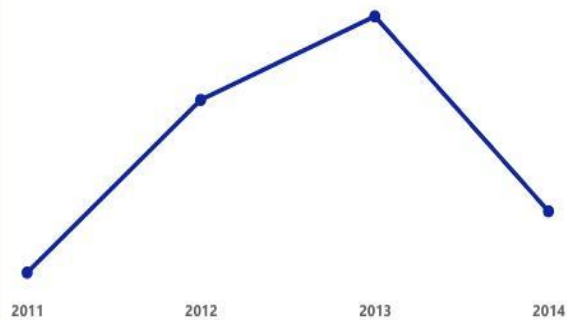
Total Quantity by Style



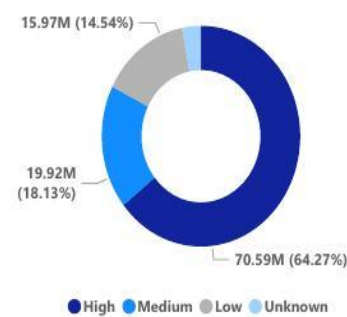
Relationship between Quantity Ordered and Revenue



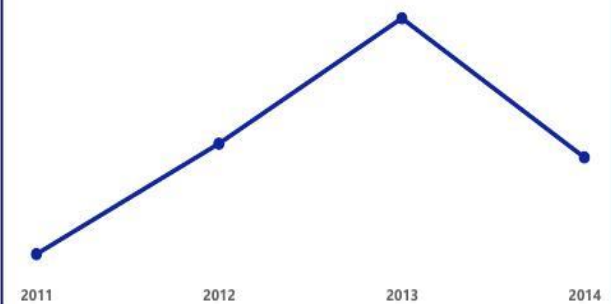
Total Revenue by Year



Total Revenue by Class



Total Quantity by Year



## COST AND PROFIT ANALYSIS

Total Cost

100.47M

Total Profit

9.37M

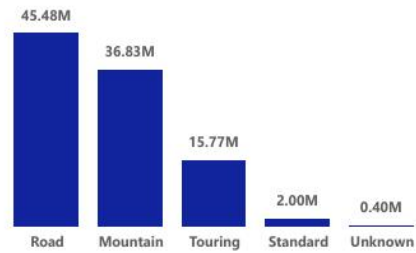
Profit Margin

0.09

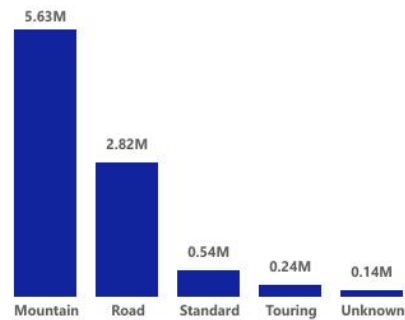
### Top 10 Product by Cost



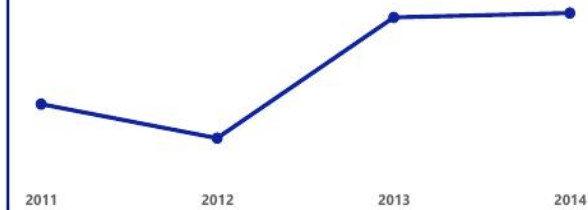
### Total Cost by Product Line



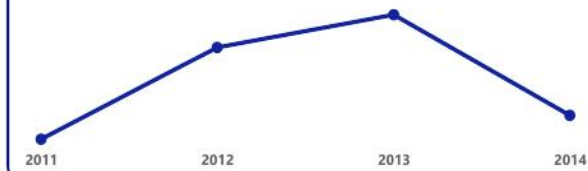
### Total Profit by Product Line



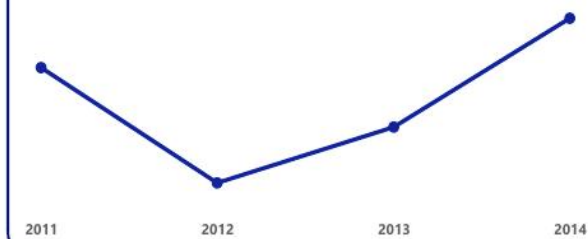
### Total Profit by Year



### Total Cost by Year



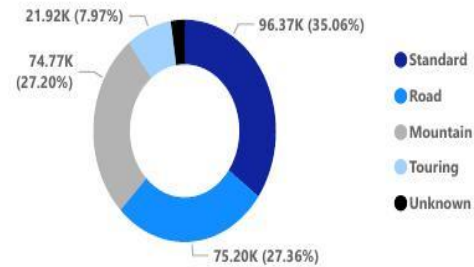
### Profit Margin by Year



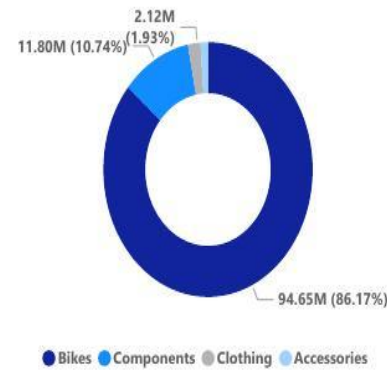


## PRODUCT PERFORMANCE

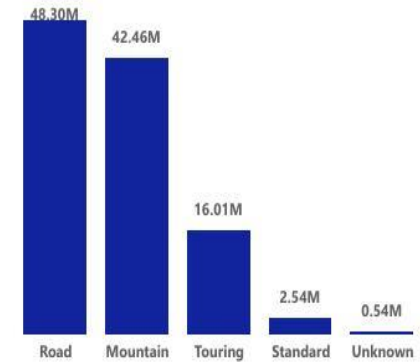
### Total Quantity by Product Line



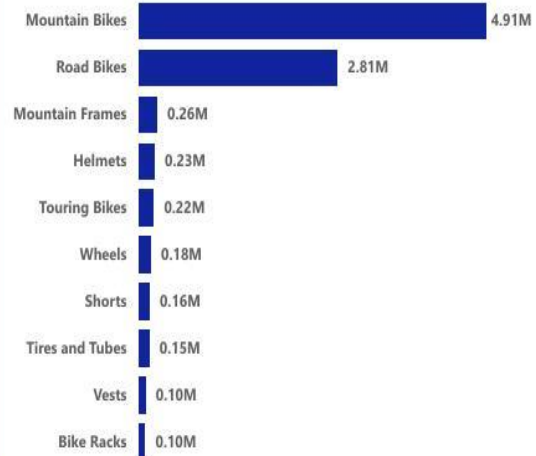
### Total Revenue by Product Category



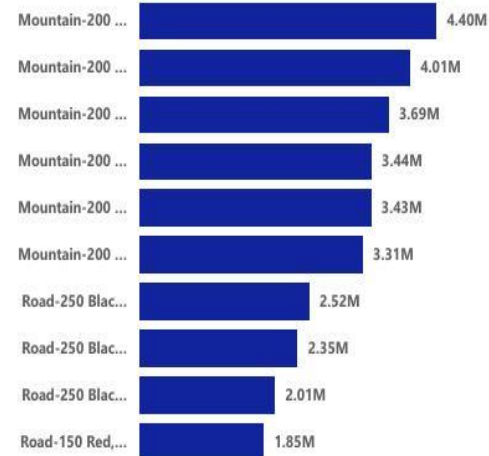
### Total Revenue by Product Line



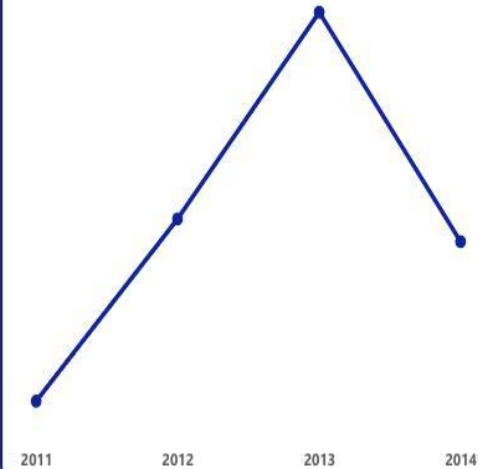
### Total Profit by Product Subcategory



### Top 10 Products by Revenue



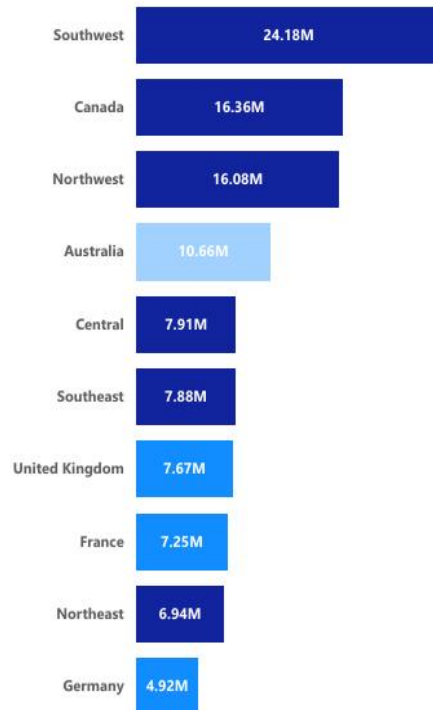
### Total Product Sold by Year



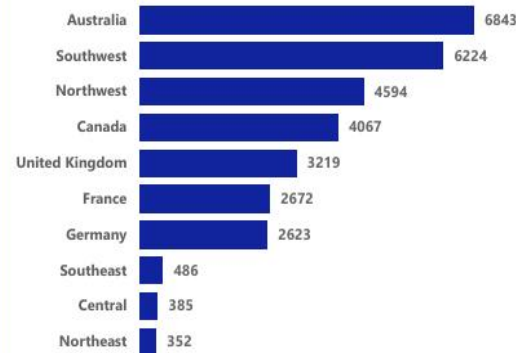
## TERRITORY ANALYSIS

### Total Revenue by Territory and Group

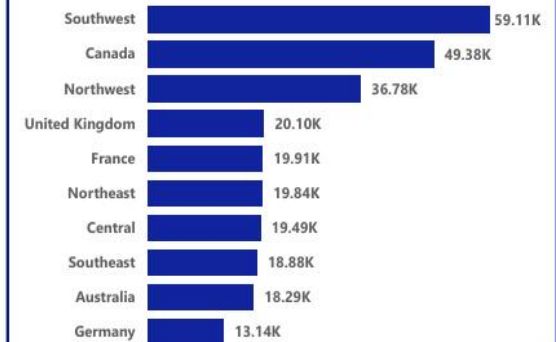
● Europe ● North America ● Pacific



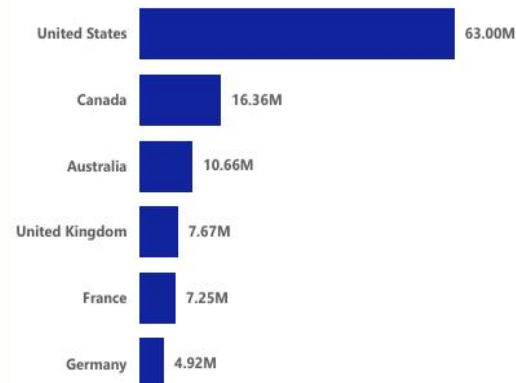
### Total Orders by Territory



### Total Product Sold by Territory



### Total Revenue by Country



### Total Profit by Territory

